KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

(DEEMED TO BE UNIVERSITY)

CENTRE FOR DISTANCE AND ONLINE EDUCATION (CDOE)

[CDOE LOGO]

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Name of the Programme: MCA **Semester: IV**

Name of the Course: Project Work **Batch: [FILL]**

Course Code: 223MCA4201 **Regulation: [FILL]**

Project Title: MaatriCare

Abstract

Pregnancy involves appointments, daily care, personal records, and communication with healthcare professionals. When information is spread across separate tools, it can be difficult to organize tasks and prepare questions for care visits. MaatriCare is a web-based pregnancy support and care-coordination application that brings these activities into one authenticated workspace.

The system provides pregnancy profile and progress information, appointment management, visit-question preparation, daily care and wellbeing tracking, a private symptom journal, milestones, educational guides and resources, healthcare summaries, emergency contacts, and reminder preferences. Users control which records are included in reports and can share selected reports through revocable, time-limited links. The application organizes user-entered information; it does not diagnose conditions, recommend treatment, or replace professional or emergency care.

The frontend uses React, TypeScript, and Vite. The backend uses Java Spring Boot REST APIs, Spring Security with JWT authentication, Jakarta validation, and Spring Data JPA. MySQL is used locally; PostgreSQL is configured for production. Optional scheduled Web Push defaults to Asia/Kolkata time and requires VAPID keys and HTTPS for live delivery. Focused backend tests and the frontend production build verify selected functionality.

Tools & Technologies Used

Frontend: React, TypeScript, Vite

Backend: Java, Spring Boot REST API

Database: MySQL (local), PostgreSQL configuration

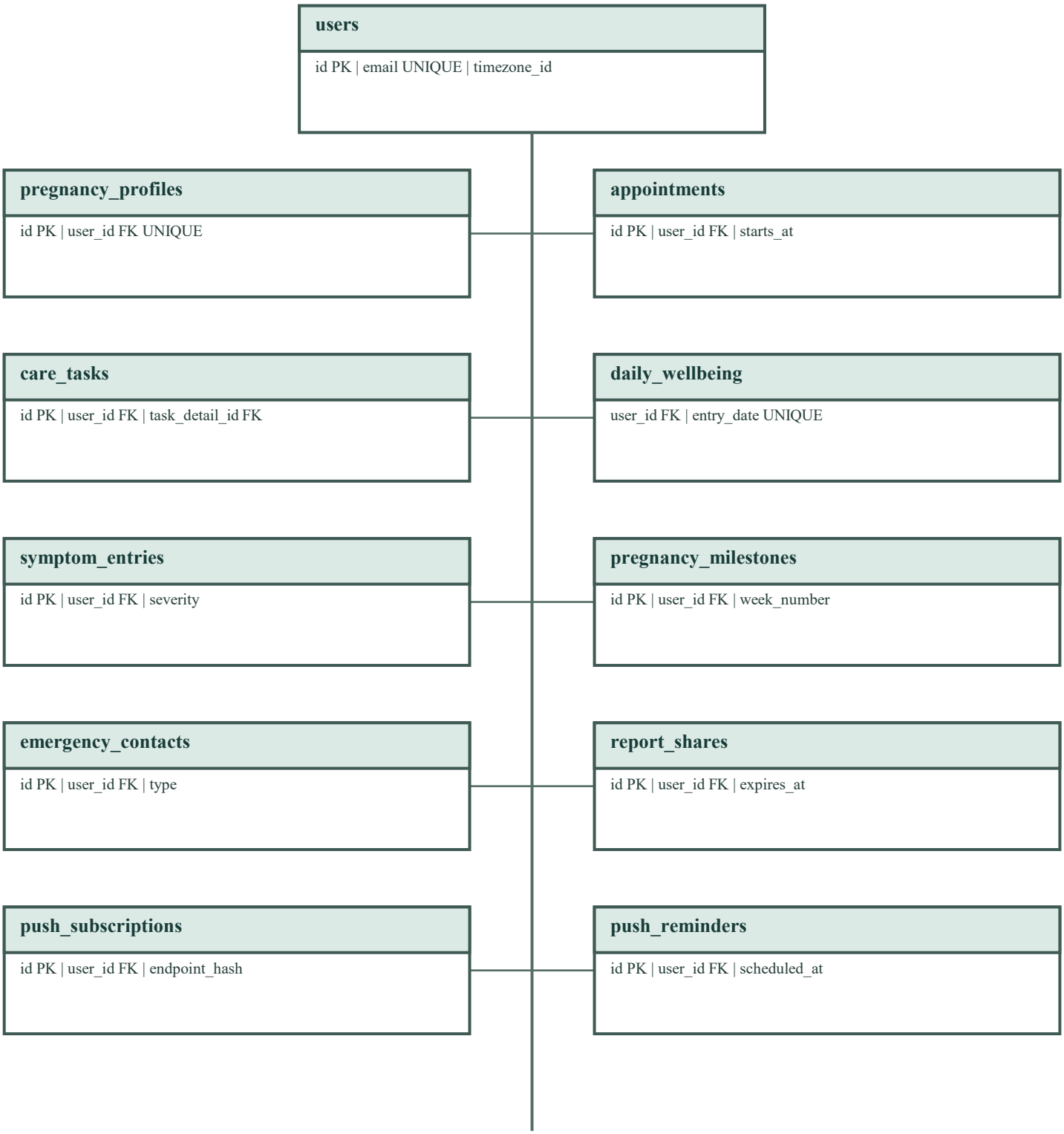
Security: Spring Security, JWT

Approval Committee

Internal Examiner	Deputy Director - CDOE	Director - CDOE

ENTITY RELATIONSHIP DIAGRAM (ER)

MaatriCare | simplified implemented schema



Relationships: one user owns one pregnancy profile and may own many appointments, tasks, wellbeing records, symptoms, milestones, shares, subscriptions, and reminders. Appointment questions reference appointments; care tasks reference task details.